

Positively charged protons attract the negatively charged electrons, holding them in orbit around the nucleus.

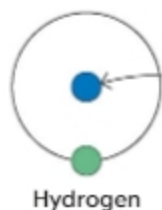


A weighty theory

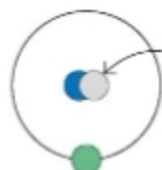
In 1808, English scientist John Dalton (left) suggested that there are different particles (or atoms) for every element and that the atoms of one element all have the same mass. German American physicist Maria Goeppert Mayer explained how the nucleus of atoms are organized, for which she won the Nobel Prize in Physics in 1963—the second of only four women to do so.

Same element, different forms

While all atoms of a particular element contain the same number of protons, some atoms of that element may have a different number of neutrons. These versions of the same element—with the same number of protons but a different number of neutrons—are called isotopes. For example, hydrogen (below) has three isotopes.

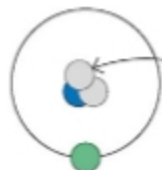


The main isotope of hydrogen has no neutrons in its nucleus.



With one extra neutron, this atom weighs twice as much as the main hydrogen isotope.

Deuterium (isotope of hydrogen)



This hydrogen isotope is three times heavier than the main hydrogen isotope.

Tritium (isotope of hydrogen)

Neutrons help keep the nucleus stable. They are attracted to the protons within the nucleus by a special form of attraction called the nuclear force.

Molecules

A molecule is a collection of neutral atoms that are chemically bonded together. The atoms may be of the same element—for example, an oxygen molecule. They can also be a collection of different atoms, as in a water molecule, which is a combination of two hydrogen atoms and one oxygen atom.

